

SMART INDIA HACKATHON 2026

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- The logo for the Smart India Hackathon (SIH) is centered. It features a stylized brain shape divided into two halves. The left half is orange and contains a circuit board pattern. The right half is green and contains binary code (0s and 1s). Below the brain is a dark blue horizontal bar with the letters 'SIH' in white. The entire logo is set against a light gray background with a hexagonal pattern.

PROBLEM

Hostile radar/comm signals hide across a huge frequency spectrum, but our receiver can only "listen" to a narrow slice at a time so it must correctly guess where and when to look, especially against emitters that hop frequencies or only appear briefly (like a scanning radar beam).

OUR SOLUTION

An ML-based adaptive scheduler that learns in real time from hits and misses dynamically deciding which band to scan next, predicting periodic/hopping emitter behavior; and reallocating search time toward active threats instead of following a blind fixed route.

OBSERVE

LEARN

PREDICT

ADAPT

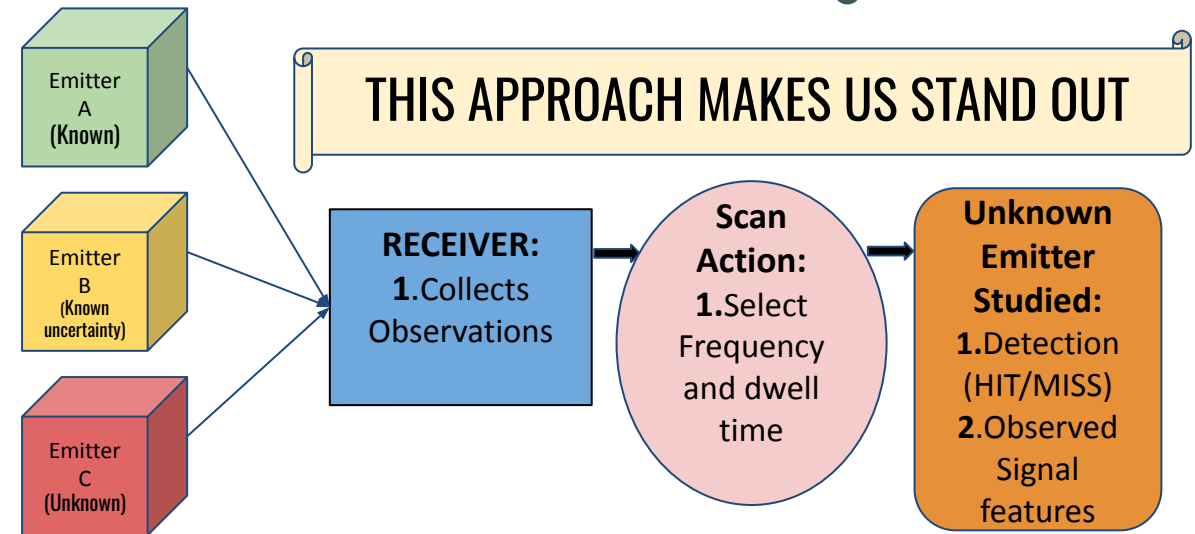
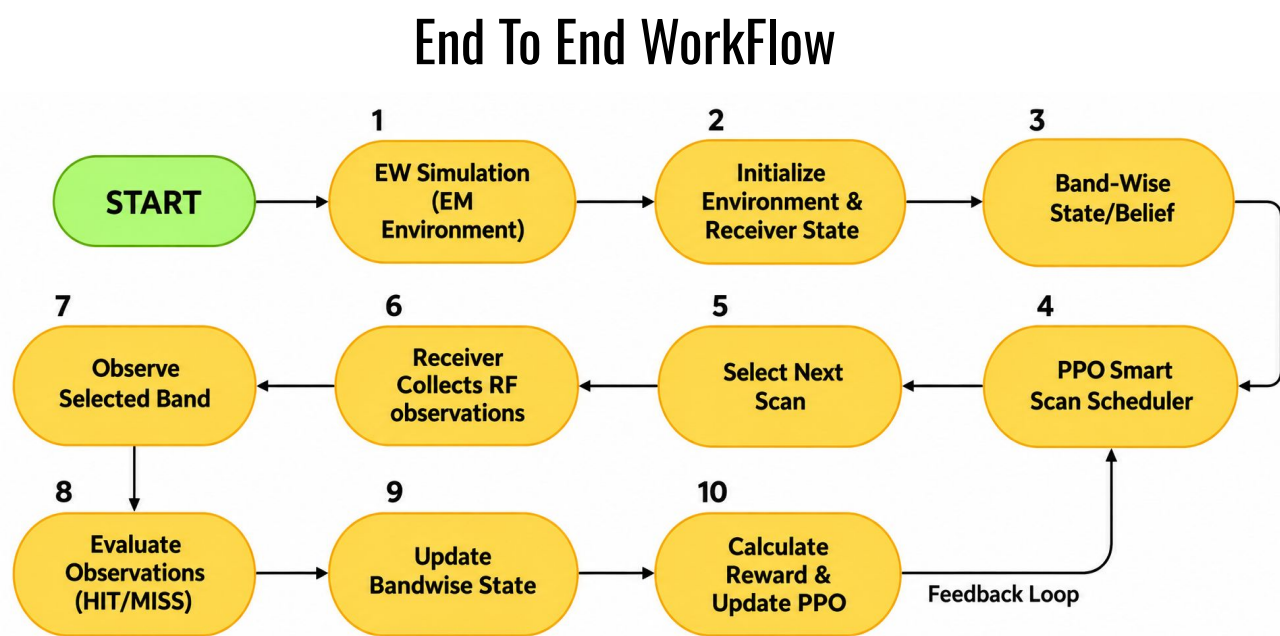
CURRENT

Existing systems use a fixed, pre-programmed sweep based only on prior intelligence - scanning band by band in the same order every time, with no adjustment based on what it's actually detecting, causing it to waste time on inactive bands and often miss new or evasive threats.

OUR USP

Unlike legacy open-loop scanners that need prior threat intelligence to work well, our system requires zero prior data it learns the environment on the fly, adapts to unknown and evasive emitters, and continuously improves interception speed and accuracy during the mission itself.

End To End WorkFlow



TECHNOLOGIES / COMPONENTS TO BE USED

Programming Language



C++



Python

ML/DL Frames



Tensor Flow



PyTorch

RL Framework



Stable Baseline 3

GUI/Visualization



Qt C++



Matplotlib/Plotly

- **Simulate** : A multi-emitter RF environment is generated as the test input.
- **Preprocessing**: PDW's from the dataset are cleaned, normalized and made consistent.
- **Decide** : The PPO agent uses the observation and uncertainty to select the next band and dwell time, balancing exploration and exploitation.
- **Act** : The receiver re-tunes to the selected band, dwell time and feeds the new observations back to the agent.

Feasibility

- 1. Technical Feasibility:** Established ML/RL technologies, EW simulation, PDW preprocessing, PPO algorithm enable prototype development.
- 2. Operational Feasibility:** Modular pipeline enables independent testing, integration and real time decision support.
- 3. Scheduling Feasibility:** A phased approach first develops and validates the baseline PPO policy, followed by LSTM/GRU integration to add temporal memory

Viability

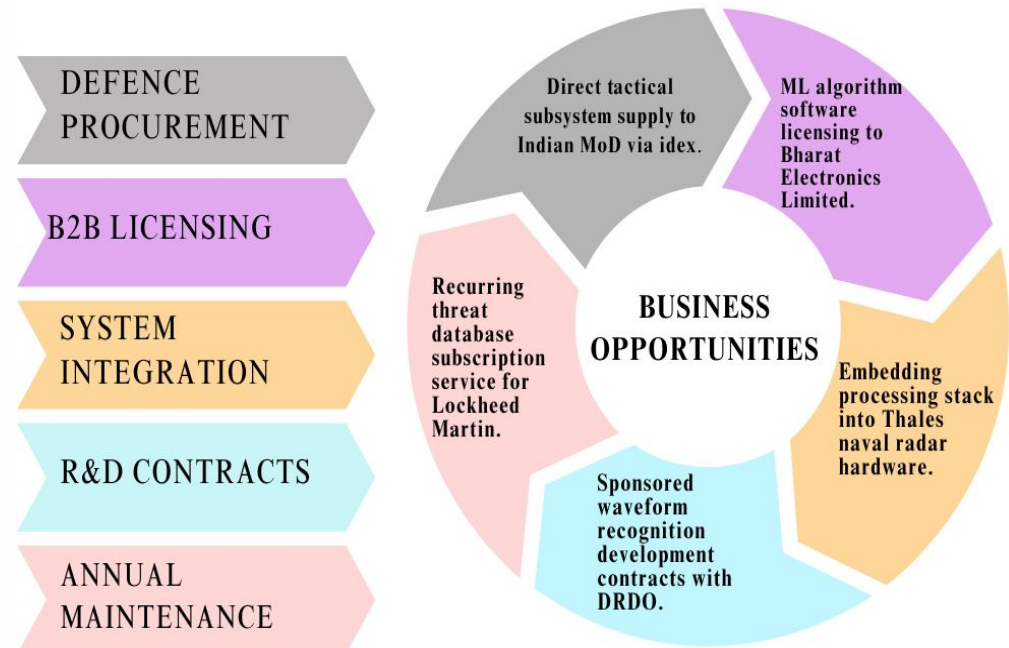
- 1. Operational Viability:** Optimizes limited receiver time by intelligently allocating frequency and dwell resources.
- 2. Deployment Viability:** Hardware-independent cognitive layer enables integration across simulated, SDR and receiver platforms.
- 3. Environmental Viability:** Adapts to unknown, intermittent and frequency-agile emitters without relying on fixed prior intelligence.

POTENTIAL CHALLENGES & RISKS

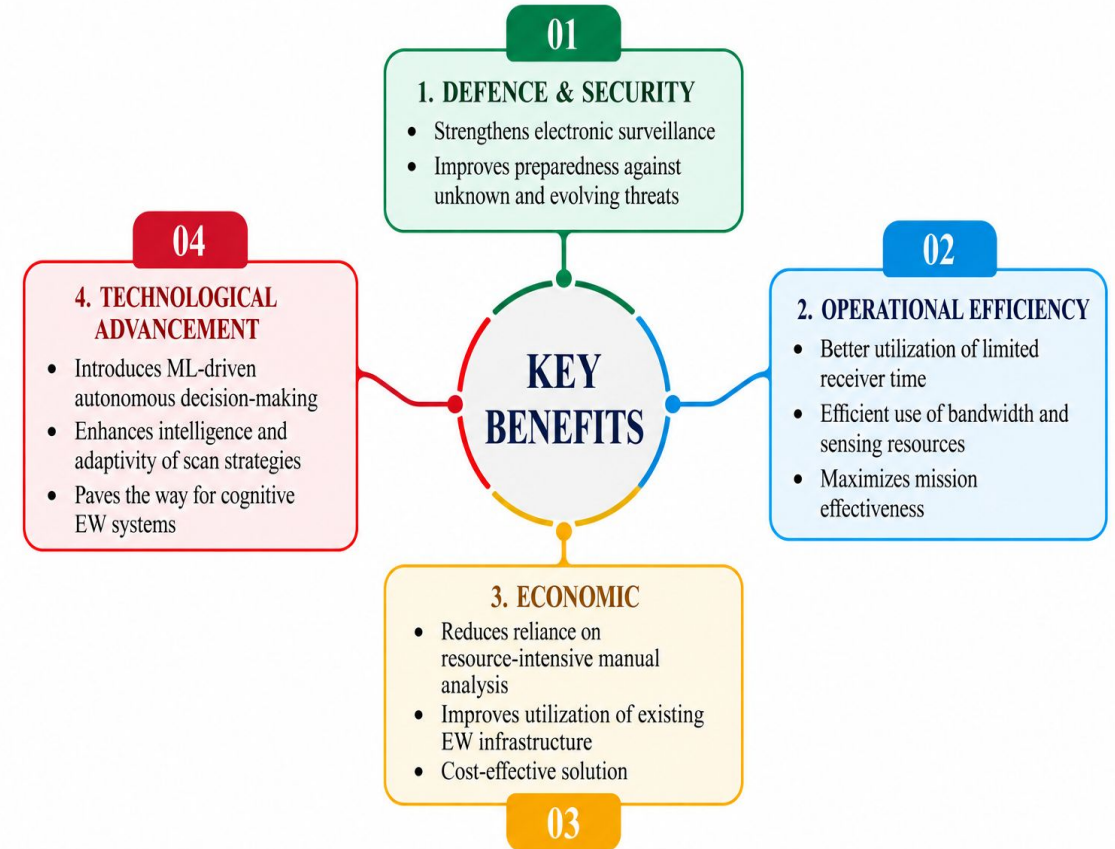
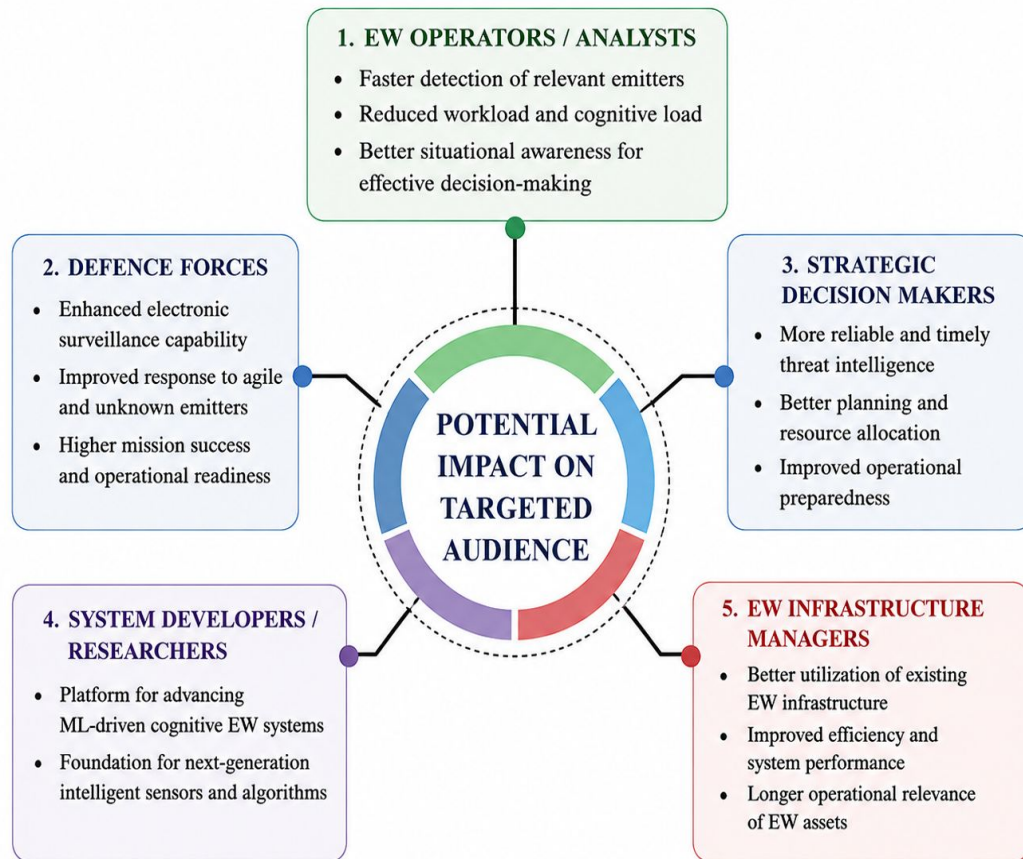
- 01 Limited Real-world EW Data**
EW data is classified and sensitive as it is related to military operations, hence real-world data is limited.
- 02 Simulation Complexity**
An unrealistic simulator can produce misleading results and poor RL performance.
- 03 RL Training Instability**
PPO may require extensive training and careful reward design to converge effectively.

STRATEGIES FOR OVERCOMING

- 01 Use Realistic Synthetic Data**
Generate diverse RF environments with varying SNR, pulse widths, PRI patterns, noise & signal overlap.
- 02 Incremental Baseline Build**
Implement step-by-step & establish a rule-based scheduler as a benchmark before RL approaches.
- 03 Careful Reward Design**
Include threat priority, tracking quality, information gain, missed detections & resource usage.



IMPACT AND BENEFITS



“India’s own smart defense shield, discovering threats early and keeping our forces one step ahead.”

[1] [V. Sekhar, G. Lechner, K. Nguyen, and S. Faulkner, "Machine Learning-based Receiver Scheduling for Electronic Support," 2025 IEEE Military Communications Conference \(MILCOM\), Los Angeles, CA, USA, pp. 1560–1565, 2025.](#)

[2] [S. Apfeld, A. Charlish, and W. Koch, "An Adaptive Receiver Search Strategy for Electronic Support," 2016 Sensor Signal Processing for Defence \(SSPD\), Edinburgh, UK, pp. 1–5, Sep. 2016.](#)

[3] [S. Apfeld, A. Charlish, and G. Ascheid, "Modelling, Learning and Prediction of Complex Radar Emitter Behaviour," 2019 18th IEEE International Conference on Machine Learning and Applications \(ICMLA\), Boca Raton, FL, USA, pp. 305–310, Dec. 2019](#)

[4] [E. Gunn, A. Hosford, R. Jones, L. Zeitler, I. Groves, and V. Nockles, "The Turing Synthetic Radar Dataset: A Dataset for Pulse Deinterleaving," arXiv preprint, arXiv:2602.03856, Jan. 2026.](#)

[5] [H. Feng, K. Jiang, and B. Tang, "Deep Multi-Intentional Inverse Reinforcement Learning for Cognitive Multi-Function Radar Inverse Cognition," Expert Systems with Applications, vol. 293, 2025](#)

[6] [M. S. J. Khan, N. L. Raichur, L. Heublein, C. Wielenberg, A. Mattick, T. Feigl, C. Mutschler, and F. Ott, "Active Sensing with Meta-Reinforcement Learning for Emitter Localization from RF Observations," arXiv preprint, arXiv:2605.12569, Jul. 2026.](#)

